

Assemble and Wrap a Burrito SOP (1x Episode: 1 Burrito)

One episode builds and wraps **one** burrito. The table begins with the **build board** in the middle, bare and dry. A stack of **wrap papers** sits at the back of the board on the **paper spot**. A **tortilla tray** holding a stack of tortillas stands in the **start zone** for the episode's config. Four **filling cups** stand in the **filling row** along the far right side, numbered **1** at the left to **4** at the right. The **cup tray** sits in the back-right corner, beyond the filling row. The **basket** sits at the front-center, between the build board and the front edge.

The order never changes: lay the paper, lay the tortilla, empty the four cups in cup order 1 to 4, fold the two ends in, roll the burrito, roll it into the paper, tuck both paper tails, then set it in the basket. No step is skipped and no step is reordered. Every fold, every roll, and every tuck happens on the build board, and the board holds one burrito at a time.

Everything rolls **toward the front edge** — the far edge of the tortilla and the far edge of the paper both come over the filling and finish underneath. Nothing is ever rolled toward the back.

The right gripper does the primary work: it takes the tortilla in Config R, empties all four filling cups, returns every empty cup, makes the right end fold, tucks the right paper tail, and takes the right half of every two-arm lift and roll. The left gripper supports: it takes the tortilla in Config L1 and L2, takes the left half of the paper, presses the tortilla down while the fillings go on, makes the left end fold, presses the burrito down while the right tail is tucked, tucks the left paper tail, and takes the left half of every two-arm lift and roll. Neither gripper reaches across the middle of the table into the other's zone.

The table is set up in one of three ways. Only the tortilla tray moves; the build board, the paper spot, the filling row, the cup tray, and the basket are in the same place in all three.

- **Config L1:** the tortilla tray is at the back-left.
- **Config L2:** the tortilla tray is at the front-left, between the basket and the left edge.
- **Config R:** the tortilla tray is at the front-right.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the tortilla tray starts at the back-left (**Config L1**), the front-left (**Config L2**), or the front-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the tortilla tray's side takes the tortilla and lays it — the left gripper in Config L1 and L2, the right gripper in Config R. No arm reaches across the table.

- **Fixed roles:** the right gripper empties and returns every cup, folds the right end, and tucks the right tail; the left gripper holds the tortilla down, folds the left end, and tucks the left tail; both grippers take every paper pick, roll, and carry. None of this changes with the config.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the paper spot at the back of the board, the build board in the middle, the filling row along the far right, the cup tray in the back-right corner, the tortilla tray in the start zone for this episode's config, and the basket at the front-center.
3. The top face of the build board is in frame from above, so the paper, the tortilla, the filling band, both end folds, and the seam can all be seen.
4. The inside of the basket is in frame from above, so the finished burrito can be seen going in.
5. Both arms are at home with grippers open.
6. The table is bare apart from the build board, the paper stack, the tortilla tray, the four filling cups, the cup tray, the basket, and robot hardware.
7. The right arm reaches all four filling cups, the cup tray, (Config R) the tortilla tray, and the right half of the build board without stretching. The left arm reaches the paper stack, (Config L1 and L2) the tortilla tray, and the left half of the build board without stretching.
8. The travel from the filling row to the build board, and from the build board to the basket, is bare and level.

Materials checklist

1. One **build board** lies flat and bare in the middle of the table, in its printed outline, wiped clean and dry. It sits on a mat that keeps it from sliding.
2. One stack of **wrap papers**, 33 cm x 33 cm square sheets, sits square and flat on the **paper spot** behind the board. The sheets pull apart one at a time and none is torn, damp, or creased.
3. One **tortilla tray** holds a stack of 28 cm flour tortillas, flat and soft, none torn, dried out, or stuck to the one below it. The tray has a rim that holds the stack while the top tortilla is

peeled off. It stands in the start zone for this episode's config, and the other two zones are bare:

- **Config L1:** back-left, clear of the paper spot
 - **Config L2:** front-left, between the basket and the left edge
 - **Config R:** front-right
4. Four **filling cups** stand in the **filling row** along the far right side, in one straight line, not touching, numbered left to right: **cup 1 rice, cup 2 beans, cup 3 protein, cup 4 salsa and cheese.**
 5. Every cup is filled to its portion line, upright, open, and wide enough at the mouth to pour without scraping. Each cup body is narrow enough for the right gripper to close across it.
 6. The **cup tray** in the back-right corner is empty and upright, beyond the filling row and clear of it.
 7. The **basket** at the front-center is empty, upright, and lined if a liner is used. Its mouth is wide enough that the finished burrito goes in flat without touching the rim.
 8. Keep the left side of the table clear apart from the tortilla tray in Config L1 and L2. The left gripper comes in from there onto the board and the paper spot.
 9. Before collection, confirm by hand that:
 - the top wrap paper pulls off the stack on its own;
 - the top tortilla peels off the stack without tearing and without lifting the one below it;
 - each cup pours clean when tipped, with nothing stuck inside it;
 - a rolled burrito lowered flat into the basket stays inside the rim.

Workspace layout

- **Paper spot:** the back of the board, directly behind the build board. Wrap papers are taken here and nowhere else.
- **Build board:** the middle of the table — the working area for the paper, the tortilla, the fillings, both folds, both rolls, and both tucks. It holds one burrito at a time and it is bare at the start and at the end.
- **Filling row:** the far right side, cups 1 to 4 left to right. Filling cups are taken here and nowhere else.
- **Cup tray:** the back-right corner, beyond the filling row. Every emptied cup goes here and nothing else ever goes in it.
- **Tortilla tray:** the start zone — back-left (**Config L1**), front-left (**Config L2**), or front-right (**Config R**). Tortillas are taken here and nowhere else.
- **Basket:** the front-center, between the build board and the front edge. The finished wrapped burrito goes here and nothing else ever goes in it.

- **Left side:** kept clear apart from the tortilla tray in Config L1 and L2, so the left gripper can come in onto the board and the paper spot.

Arm assignments

- **Right gripper:** in Config R, takes the top tortilla and lays it; picks and empties all four filling cups in cup order, returns every empty cup to the cup tray, makes the right end fold, tucks the right paper tail, and takes the right half of every two-arm lift, roll, and carry.
- **Left gripper:** in Config L1 and L2, takes the top tortilla and lays it; takes the left half of the wrap paper, presses the tortilla down through the whole filling step, makes the left end fold, presses the wrapped burrito down while the right tail is tucked, tucks the left paper tail, and takes the left half of every two-arm lift, roll, and carry.

Vocabulary

- **Build board:** the flat board in the middle of the table that the burrito is built on. It never leaves its outline and is never lifted.
- **Wrap paper:** one square sheet from the paper spot. One sheet per burrito.
- **Tortilla:** one flour tortilla from the tortilla tray. One tortilla per burrito.
- **Start zone:** where the tortilla tray stands at the start of the episode — back-left (**Config L1**), front-left (**Config L2**), or front-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Filling cup:** one pre-portioned cup from the filling row. Four cups are emptied in one episode.
- **Cup order:** cup 1 first, then cup 2, then cup 3, then cup 4. The order never changes.
- **Near edge:** the edge of the paper or the tortilla closest to the front edge of the table.
- **Far edge:** the edge of the paper or the tortilla farthest from the front edge of the table.
- **Left end / right end:** the two sides of the tortilla, to the left and to the right of the filling band.
- **Filling band:** the strip the fillings are laid in — a band running left to right across the tortilla, sitting between the middle of the tortilla and its far edge, about a third of the tortilla's width long, with bare tortilla left at both ends of the band and at the near and far edges.
- **Inside the band:** all four fillings land in that one strip and nowhere else. Filling that spreads out wide, or lands near an edge, squeezes out during the roll.
- **Emptied:** the cup has been tipped until nothing is left in it, with no shaking and no scraping against the tortilla.
- **End fold:** the left end or the right end of the tortilla brought inward over the filling and pressed flat, so the folded edge lands just clear of that end of the filling band, without covering the band.
- **Fold stands:** the end fold stays down on its own with no gripper on it.

- **Fold pops open:** the end fold springs back up after the gripper lets go.
- **Tuck back:** with the far edge already brought over the filling, both grippers pull it back toward the far side a little, so the filling packs into a tight cylinder.
- **Roll:** press down with both grippers and turn the bundle toward the front edge in one continuous motion. A roll turns the bundle over; it never drags or slides it.
- **Burrito:** the rolled tortilla before the paper goes on — a closed cylinder, the same thickness along its whole length, seam down, with no filling showing at either end.
- **Seam:** the line where the tortilla's near edge finishes. The seam always finishes underneath.
- **Paper tail:** the paper sticking out past each end of the burrito after the paper roll, before it is tucked.
- **Tail tuck:** the paper tail flattened against the board, folded in against the end of the burrito, then folded under it.
- **Wrapped burrito:** the finished result — the burrito fully covered by paper, seam down, both tails folded flat underneath, and holding closed with no gripper on it.
- **Level carry:** both grippers hold the burrito at the same height and move together, so it stays flat and the tucks stay underneath.

Steps

Run Steps 1–8 in order on the one build board, then end the episode with Step 9. Step 2 depends on where the tortilla tray is: in Config L1 and L2 the **left gripper** takes the tortilla and lays it; in Config R the **right gripper** does. Every other line is the same in all three configs.

Step 1: Lay a wrap paper on the build board

Goal: one wrap paper lies flat and square in the middle of the build board.

- With the **left gripper**, pinch the near-left corner of the top paper. With the **right gripper**, pinch its near-right corner.
- Take the top sheet only.
- Both grippers lift it straight up, clear of the stack, then carry it level to the board.
- Both grippers lower it flat in the middle of the board, square, with its near and far edges running left to right across the board.
- Both grippers let go once the sheet is resting, then lift clear.

Check: one sheet only, lying flat and square in the middle of the board, no corner curled under, and no part hanging off the board. If two sheets came up, both grippers lift the top one off and lay it back on the paper spot. If the sheet is crooked or curled, both grippers pick it up by the near corners and lay it again.

Expected state: one flat, square sheet on the board, the paper stack undisturbed, and both grippers empty.

Step 2: Lay a tortilla on the paper

Goal: one tortilla lies flat and open on the far half of the wrap paper.

Look where the tortilla tray is before reaching for the tortilla.

- **IF the tortilla tray is at the back-left (Config L1):** with the **left gripper**, pinch the edge of the top tortilla, peel it straight up, clear of the stack, and carry it level forward and to the right, to the paper.
- **IF the tortilla tray is at the front-left (Config L2):** with the **left gripper**, pinch the edge of the top tortilla, peel it straight up, clear of the stack, and carry it level back and to the right, to the paper.
- **IF the tortilla tray is at the front-right (Config R):** with the **right gripper**, pinch the edge of the top tortilla, peel it straight up, clear of the stack, and carry it level back and to the left, to the paper.

Then, in all three:

- Take the top tortilla only. If a second one lifts with it, lay both back in the tray with the same gripper and take the top one again.
- Lower it flat, centered side to side on the paper, with its far edge just inside the paper's far edge, so there is bare paper all the way around it.
- Open the gripper, let go, and lift clear. The other gripper stays clear.

Check: the tortilla lies flat and open on the far half of the paper, with no folded edge, no curled edge, no tear, and nothing hanging off the paper. If an edge is folded under, flatten it with the same gripper. If the tortilla is torn, lift it off with the same gripper, set it aside off the board, and take a fresh one.

Expected state: one tortilla flat on the far half of one paper, both grippers empty, and the four cups still standing in the filling row.

Step 3: Empty the four filling cups onto the tortilla

Goal: all four fillings lie stacked in one filling band on the tortilla, and all four empty cups stand in the cup tray.

- With the **left gripper**, closed, press down on the near edge of the tortilla and hold it there for the whole of Step 3, so the tortilla and the paper cannot slide.
- Empty the cups in **cup order: 1, then 2, then 3, then 4**. Handle one cup at a time.
- Every empty cup is returned to the cup tray before the next cup is picked up.

3.1 Empty a cup onto the tortilla

- With the **right gripper**, close across the side of the next cup in order and lift it straight up, clear of the row.
- With the **right gripper**, carry it over the tortilla and lower it until the cup mouth is just above the tortilla, close enough that the filling drops straight down and does not scatter.
- With the **right gripper**, tip the cup away from the front edge and move it slowly from the left end of the filling band to the right end, so the filling lands as one even strip inside the band.
- Keep tipping with the **right gripper** until the cup is empty. Do not shake the cup and do not scrape it against the tortilla.

3.2 Return the empty cup

- With the **right gripper**, carry the empty cup to the **cup tray** in the back-right corner.
- Set it down upright in the tray, open the **right gripper**, and lift clear.
- Go back to 3.1 for the next cup in order.

Check: after each cup, that filling lies inside the band on top of the ones before it, no filling has landed near a tortilla edge, and the cup stands empty in the cup tray. If filling lands outside the band, push it back into the band with the **right gripper** before the next cup. If a cup is not empty, pick it back up with the **right gripper** and keep tipping.

Expected state: four fillings stacked in one band, bare tortilla all around the band, four empty cups standing in the cup tray, the filling row empty, and the left gripper still pressing the tortilla down.

Step 4: Fold the two ends in

Goal: both ends of the tortilla are folded inward over the filling, lying flat, and both folds stand on their own when the roll begins.

4.1 Fold the left end

- Open the **left gripper** and lift it off the tortilla's near edge.
- With the **left gripper**, pinch the middle of the tortilla's **left edge** and lift it clear of the board.
- With the **left gripper**, bring it inward over the filling until the folded edge lands just clear of the left end of the filling band, and press it down flat.
- Open the **left gripper** and lift clear.

4.2 Fold the right end

- With the **left gripper**, closed, press down on the left fold so it cannot pop open.
- With the **right gripper**, pinch the middle of the tortilla's **right edge**, lift it, and bring it inward over the filling until the folded edge lands just clear of the right end of the band.
- With the **right gripper**, press it down flat, then open and lift clear.
- Open the **left gripper** and lift it off the left fold.

Check: both folded edges run front to back, sit parallel to each other, each land just clear of its end of the band, and neither fold covers the band. Both folds stand with no gripper on them. If a fold pops open, press it down again with the gripper that made it.

Expected state: both ends folded in and standing, the filling still inside the band, and both grippers clear of the board.

Step 5: Roll the burrito

Goal: one closed burrito, the same thickness along its whole length, seam down, with no filling showing at either end.

5.1 Bring the far edge over the filling

- With the **left gripper**, pinch the tortilla's **far edge** to the left of center. With the **right gripper**, pinch it to the right of center. Hold both at the same height.
- Both grippers lift the far edge together, carry it over the filling band, and lower it onto the tortilla just on the near side of the band.

5.2 Tuck back against the filling

- With both grippers still holding the edge, pull it back toward the far side a little, so the filling packs into a tight cylinder.
- Keep both grippers level so the cylinder is the same thickness at both ends.

5.3 Roll toward the front edge

- Press down with both grippers and roll the bundle toward the front edge in one continuous motion, until the tortilla's near edge finishes underneath.
- Keep both end folds tucked in as the bundle turns. Do not let an end fold ride out.
- Roll the bundle. Never drag or slide it across the board.
- Open both grippers and lift clear.

Check: the burrito is closed, seam down, the same thickness along its whole length, with both ends closed and no filling showing. If an end fold has ridden out, unroll the burrito with both grippers, press that fold back down, and roll again. If the roll is loose, unroll it with both grippers and roll again from 5.2 with a firmer tuck back.

Expected state: one closed burrito lying seam down on the wrap paper, both grippers clear, and no filling on the board.

Step 6: Roll the burrito into the paper

Goal: the burrito is fully covered by the paper, seam down, with a paper tail at each end.

- Do not start this step until the burrito is a closed cylinder. Paper put on an open roll cannot be fixed.

- With the **left gripper**, pinch the paper's **far edge** to the left of center. With the **right gripper**, pinch it to the right of center.
- Both grippers lift the far edge together, bring it over the burrito, and lower it onto the paper just on the near side of the burrito, pulling it snug against the burrito.
- Press down with both grippers and roll the burrito toward the front edge in one continuous motion, until the paper's near edge finishes underneath.
- Press the wrapped burrito down flat with both grippers so it cannot unroll, then lift clear.

Check: no tortilla shows, the paper lies snug with no loose pocket of air, the seam is down, and a paper tail sticks out at each end. If the paper is loose, hold the burrito down with the **left gripper**, pull the paper snug with the **right gripper**, and press it down again.

Expected state: the burrito wrapped in paper, seam down, with one paper tail at each end, still on the board.

Step 7: Tuck the two paper tails

Goal: both paper tails are folded flat under the burrito, and the wrapped burrito holds closed on its own.

7.1 Tuck the left tail

- With the **right gripper**, closed, press down on the middle of the wrapped burrito and hold it there.
- With the **left gripper**, flatten the left tail against the board, fold it in against the end of the burrito, then fold it under the burrito.
- Open the **left gripper** and lift clear.

7.2 Tuck the right tail

- With the **left gripper**, closed, press down on the middle of the wrapped burrito and hold it there.
- With the **right gripper**, flatten the right tail against the board, fold it in against the end of the burrito, then fold it under the burrito.
- Open the **right gripper** and lift clear, then open the **left gripper** and lift clear.

Check: both tails lie flat underneath the burrito, no paper stands up or hangs loose, and the wrapped burrito stays closed with no gripper on it. If a tail lifts, press the burrito down with one gripper and fold that tail under again with the other.

Expected state: one wrapped burrito on the board, seam and both tucks down, holding closed on its own.

Step 8: Put the burrito in the basket

Goal: the wrapped burrito lies flat in the middle of the basket, seam and tucks down.

- With the **left gripper**, close on the burrito near its left end. With the **right gripper**, close on it near its right end.
- Both grippers lift it together and level, clear of the board, keeping the tucks underneath.
- Never carry the wrapped burrito on one arm.
- Both grippers carry it level to the basket and lower it flat into the middle, with its long side running left to right.
- Both grippers let go together once it is resting, then lift clear.

Check: the burrito lies flat in the middle of the basket, seam and tucks down, fully inside the rim, and still closed. If it lands on the rim or on its side, pick it up with both grippers and lower it again.

Expected state: the burrito in the basket, the build board bare, and both grippers empty.

Step 9: End the episode

Goal: recording ends with one wrapped burrito in the basket and the board bare.

1. Confirm the end state:
 - one wrapped burrito lies flat in the basket, seam and tucks down, fully inside the rim;
 - no tortilla shows through the paper and no paper stands up;
 - the build board is bare, with no filling, no paper, and no tortilla left on it;
 - four empty cups stand in the cup tray and the filling row is empty;
 - the paper stack and the tortilla tray are still on their spots.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

After each episode

1. Take the wrapped burrito out of the basket and set it aside off the table.
2. Take the four empty cups out of the cup tray, refill them to their portion lines, and stand them back in the filling row in order: cup 1 rice at the left, then beans, protein, salsa and cheese at the right.
3. Wipe the build board clean and dry. Brush off any filling, paper scraps, or tortilla crumbs.
4. Square the paper stack on the paper spot, and square the tortilla stack in the tortilla tray. Stand the tray in the start zone for the next episode's config — back-left (Config L1), front-left (Config L2), or front-right (Config R) — and leave the other two zones bare.
5. Check the basket is empty, upright, and clean, and re-line it if a liner is used.

6. Run both Setup checklists before the next episode.

At the end of the session

1. Inspect the supplies. Throw out tortillas that have dried out, cracked, or torn. Throw out wrap papers that are creased, damp, or torn. Replace a cup that is cracked or no longer pours clean.
2. Wash and dry the four cups, the cup tray, and the basket.
3. Leave the build board bare and dry in its outline, the paper stack square on the paper spot, the tortilla tray covered in the start zone for the next session's config, the four cups empty and upright in the filling row, and the cup tray and basket empty.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config R (tortilla tray at the front-right, the right gripper taking the tortilla). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the tortilla tray is not in the start zone for the config; a gripper reaches across the table for the tortilla; or the wrong IF line is followed.

- **SOP rule broken:** the start position and the same-side rule (the gripper on the tortilla tray's side takes the tortilla and lays it — the left gripper in Config L1 and L2, the right gripper in Config R; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the tortilla tray is before the first reach for it, then follow that config's IF line through Step 2.

Violation: Wrong paper pick

- **Visible cue:** two or more sheets leave the paper spot in one pick, a sheet is taken from somewhere other than the paper spot, or the sheet is dragged across the table instead of being lifted clear and carried.
- **SOP rule broken:** Step 1 (both grippers pinch the near corners of the top sheet only, lift it clear of the stack, and carry it level to the board).
- **Coaching note:** two corners, lift straight up, and check one sheet came free before you carry it.

Violation: Paper laid crooked, curled, or off the board

- **Visible cue:** the sheet lands out of square with the front edge, with a corner curled under or wrinkled, off the middle of the board, or with part of it hanging off the board, and the episode moves on.
- **SOP rule broken:** Step 1 (both grippers lower the sheet flat and square in the middle of the board, with no corner curled under and no part hanging off).
- **Coaching note:** lower both corners together and let go only once the sheet is resting flat.

Violation: Wrong tortilla pick

- **Visible cue:** two or more tortillas leave the tray in one pick, the stack lifts with the pick, a tortilla is taken from somewhere other than the tortilla tray, or the tortilla tears as it comes off and is used anyway.
- **SOP rule broken:** Step 2 (the right gripper peels the top tortilla only, straight up and clear of the stack, and a torn tortilla is set aside for a fresh one).
- **Coaching note:** peel from the edge, straight up, one tortilla. If it tears, take a new one.

Violation: Tortilla laid badly

- **Visible cue:** the tortilla lands with a folded or curled edge, off-center on the paper, hanging off the paper, or on the near half instead of the far half, and the episode moves on.
- **SOP rule broken:** Step 2 (the right gripper lowers the tortilla flat and centered on the far half of the paper, with bare paper all the way around it).
- **Coaching note:** far half, centered, flat. Flatten any folded edge before you let go.

Violation: Tortilla not held during the filling

- **Visible cue:** the left gripper is off the tortilla's near edge while the right gripper empties a cup, and the tortilla or the paper slides, lifts, or is chased across the board.
- **SOP rule broken:** Step 3 (the left gripper presses down on the near edge of the tortilla and holds it there for the whole of Step 3).
- **Coaching note:** left gripper down first, and leave it there until both fillings and folds are done with it.

Violation: Cups emptied out of order or a cup skipped

- **Visible cue:** the right gripper empties the cups in some order other than 1, 2, 3, 4, or one of the four cups is never emptied and the episode moves on to the folds.
- **SOP rule broken:** Step 3 (empty the cups in cup order 1, then 2, then 3, then 4, one cup at a time).
- **Coaching note:** left to right along the row, all four. Say the cup number before you pick it up.

Violation: More than one cup handled at a time

- **Visible cue:** the right gripper lifts a second cup before the last one is standing in the cup tray, or two cups are held or carried at once.
- **SOP rule broken:** Step 3 (handle one cup at a time, and return every empty cup to the cup tray before picking up the next).
- **Coaching note:** one cup out, one cup back, then the next.

Violation: Filling poured badly

- **Visible cue:** the right gripper tips a cup from high above the tortilla so the filling scatters, shakes the cup, or scrapes it against the tortilla.
- **SOP rule broken:** Step 3 (lower the cup mouth to just above the tortilla and tip it slowly, without shaking and without scraping).
- **Coaching note:** get the mouth down close, then tip slow and travel along the band.

Violation: Filling outside the band

- **Visible cue:** filling lands spread wide across the tortilla, near an edge, or outside the band, and the episode moves on without it being pushed back in.
- **SOP rule broken:** Step 3 (all four fillings land inside the one filling band, with bare tortilla left at both ends and at the near and far edges).
- **Coaching note:** keep the pour inside the band. Filling near an edge squeezes out in the roll and cannot be recovered.

Violation: Cup not emptied or not returned

- **Visible cue:** the right gripper sets a cup down with filling still in it, leaves an empty cup on the board, on the table, or back in the filling row, or drops it into the basket.
- **SOP rule broken:** Step 3 (keep tipping until the cup is empty, then set it upright in the cup tray).
- **Coaching note:** empty it fully, then stand it in the cup tray. Nothing else goes in that tray.

Violation: End fold missed or left popped open

- **Visible cue:** the episode reaches the roll with one or both ends unfolded, or with a fold that sprang back up and was not pressed down again.
- **SOP rule broken:** Step 4 (both ends are folded inward, pressed flat, and standing on their own before the roll begins).
- **Coaching note:** press the fold, let go, and watch it. If it lifts, press it again before you move on.

Violation: End fold placed wrong

- **Visible cue:** a fold lands short of the filling band and leaves the end open, or it is brought so far in that the folded edge covers the band.
- **SOP rule broken:** Step 4 (bring each end in until the folded edge lands just clear of its end of the filling band, without covering the band).
- **Coaching note:** just past the end of the filling, no further. Aim for the bare strip beside the band.

Violation: Far edge not tucked back

- **Visible cue:** both grippers bring the far edge over the filling and go straight into the roll, with no pull back toward the far side, and the finished burrito is loose or uneven in thickness.
- **SOP rule broken:** Step 5 (with both grippers still holding the far edge, pull it back toward the far side so the filling packs into a tight cylinder).
- **Coaching note:** over, then back. The pull back is what makes it tight.

Violation: Burrito dragged instead of rolled

- **Visible cue:** the bundle slides or is pushed across the board instead of turning over, or the roll stops and restarts part way instead of running in one motion.
- **SOP rule broken:** Step 5 (press down with both grippers and roll the bundle toward the front edge in one continuous motion).
- **Coaching note:** press and turn it over. If it skids, you are pushing, not rolling.

Violation: Burrito left open

- **Visible cue:** the roll finishes with the seam on top or on the side, an end fold ridden out, filling showing at an end, or the cylinder much fatter at one end, and the episode moves on

to the paper.

- **SOP rule broken:** Steps 5 and 6 (the roll ends as a closed cylinder, seam down, even in thickness, with no filling showing, and the paper does not go on until it is).
- **Coaching note:** check the roll before the paper. Once it is wrapped, an open roll cannot be fixed.

Violation: Paper not rolled home or left loose

- **Visible cue:** the paper roll stops before the paper's near edge finishes underneath, tortilla still shows through, or the paper stands off the burrito in a loose pocket of air.
- **SOP rule broken:** Step 6 (roll the burrito toward the front edge until the paper's near edge finishes underneath, snug, with no tortilla showing).
- **Coaching note:** pull the paper snug against the burrito before you roll, and roll it all the way under.

Violation: Paper tail not tucked

- **Visible cue:** the episode ends, or the burrito is carried, with a tail standing up, hanging loose, or folded in against the end but never folded under.
- **SOP rule broken:** Step 7 (flatten each tail, fold it in against the end of the burrito, then fold it under the burrito).
- **Coaching note:** flatten, fold in, fold under. All three, both ends.

Violation: Burrito not held while a tail is tucked

- **Visible cue:** neither gripper is pressing the wrapped burrito down while the other folds a tail, and the burrito rolls, shifts, or unwinds on the board.
- **SOP rule broken:** Step 7 (one gripper presses down on the middle of the wrapped burrito while the other tucks that tail).
- **Coaching note:** hold it down first, then tuck. One gripper always stays on it.

Violation: Burrito carried on one arm

- **Visible cue:** one gripper lifts and carries the wrapped burrito to the basket on its own, or the two grippers lift at different heights and the burrito tilts on the way.
- **SOP rule broken:** Step 8 (both grippers close near the two ends, lift together and level, and carry it level to the basket).
- **Coaching note:** two grippers, same height, short level move. A one-arm carry unwinds the paper.

Violation: Burrito dropped into the basket

- **Visible cue:** the grippers open above the basket rim and the burrito falls in, lands on the rim, lands on its side, or lands with the seam or a tuck facing up.

- **SOP rule broken:** Step 8 (lower it flat into the middle of the basket, seam and tucks down, and let go only once it is resting).
- **Coaching note:** take it down into the basket and set it down. Let go only when it is already lying flat.

Violation: Work done out of order

- **Visible cue:** the phases run out of sequence — a filling goes on before the tortilla is laid, an end is folded before all four cups are empty, the paper goes on before the burrito is rolled, or a tail is tucked before the paper roll finishes.
- **SOP rule broken:** Steps 1–8 (paper, tortilla, four fillings in cup order, both end folds, the roll, the paper roll, both tucks, then the basket, in that order).
- **Coaching note:** finish the step before you start the next one. Say the step name before you move.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the left gripper emptying a cup, taking a tortilla, or making the right end fold, or the right gripper making the left end fold or tucking the left tail.
- **SOP rule broken:** Steps 1–8 (the right gripper takes the tortilla, empties and returns every cup, folds the right end, and tucks the right tail; the left gripper holds the tortilla down, folds the left end, and tucks the left tail).
- **Coaching note:** right gripper works the right side and the cups, left gripper holds and works the left side. It does not swap.

Violation: Item dropped or knocked off its spot

- **Visible cue:** either gripper drops a tortilla, a paper, a cup, or the burrito on the table or the floor, spills filling off the tortilla onto the board or the table, or tips, pushes, or drags the build board, the cup tray, the tortilla tray, or the basket off its spot.
- **SOP rule broken:** Steps 1–8 (the board, the trays, and the basket stay on their spots, and every carry stays level and low over the table).
- **Coaching note:** work slower and lower over the table, and keep each carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the wrapped burrito and the bare board are confirmed, an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 9 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- A tortilla arrives already cracked or torn through, so it cannot be rolled closed.
- Wrap papers arrive stuck together or damp, so a single sheet cannot be taken.
- A filling cup arrives under-filled, over-filled, or with the filling stuck solid inside it.
- The basket is cracked or will not stand upright, so the burrito cannot be set down inside the rim.

Annotation subtasks (from SOP)

1. Lay one wrap paper flat and square on the build board with both grippers
2. Peel the top tortilla and lay it on the far half of the paper
3. Press the tortilla down with the left gripper
4. Empty one filling cup into the filling band with the right gripper
5. Return the empty cup to the cup tray
6. Repeat the empty and return for all four cups in order
7. Fold the left end in with the left gripper
8. Fold the right end in with the right gripper
9. Bring the far edge over, tuck back, and roll the burrito toward the front edge
10. Bring the paper's far edge over and roll the burrito into the paper
11. Tuck the left paper tail, then the right paper tail
12. Lift the wrapped burrito with both grippers and lower it into the basket
13. Confirm the end state, return both arms home, and end the episode